

 Skill Verse — Testing Quick Reference Guide

4 Core Levels of Software Testing: Commands, Tools & Scope

CSE 327
Reference

 Testing Commands & Usage

COMMAND	TOOL USED	WHAT IT DOES
<code>npm run test</code>	Vitest	Runs all Unit & Integration tests once in the terminal. Shows quick pass/fail results.
<code>npm run test:watch</code>	Vitest Watch	Runs Unit tests in live mode. Automatically re-tests affected files instantly when you press <code>Ctrl+S</code> .
<code>npm run test:e2e</code>	Playwright	Runs System & UAT browser tests invisibly in the background across Chrome/Safari.
<code>npm run test:e2e:ui</code>	Playwright UI	Opens an interactive visual window to step through browser tests, time-travel, inspect network calls, and debug errors.

 Testing Tools & Their Roles

Vitest Fast Vite-native test runner for Unit and Integration tests. Replaces legacy Jest with instant ESM execution.	Playwright End-to-End browser automation framework. Simulates real users interacting with Chrome/Firefox/Safari browsers.
React Testing Library Renders React components in memory to test button clicks, inputs, and DOM output as users see them.	jsdom In-memory browser DOM emulator that allows fast unit testing without needing to launch a full web browser window.

 What We Are Testing (Very Shortly)

TESTING LEVEL	TARGET FILES	WHAT IS VERIFIED (SHORT SUMMARY)
Level 1: Unit	<code>nbutton</code> , <code>ncard</code> , <code>ninput</code> , <code>progress</code> , <code>file-parser</code>	UI rendering, variant colors, click handlers, text input changes, 100% progress indicators, and file byte calculations.

TESTING LEVEL	TARGET FILES	WHAT IS VERIFIED (SHORT SUMMARY)
Level 2: Integration	<code>auth-context</code> , <code>ai-task-prompt</code>	React components interacting with Auth Context/localStorage, and AI task urgency calculations with workload context aggregation.
Level 3: System	<code>system-middleware</code> , <code>landing-navigation</code>	Next.js Middleware route security (blocking unauthenticated access to <code>/learner</code> and <code>/educator</code>) and landing page links.
Level 4: UAT	<code>uat-user-journey</code> , <code>dashboard-loading</code>	End-to-end user onboarding journey, role selection UI, and login form control integrity.